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United States Patent	12418139
Kind Code	B2
Date of Patent	September 16, 2025
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Rubber plug unit and connector

Abstract

A rubber plug unit (**10, 10**A, **10**B) is provided with a terminal (**13, 13**A, **14, 14**A), a wire (**11**) and a rubber plug (**12, 12**A, **12**B) to be held in close contact with the outer periphery of the wire (**11**). The wire (**11**) includes an electrically conductive portion (**22**) to be connected to the terminal (**13, 13**A, **14, 14**A). The rubber plug (**12, 12**A, **12**B) includes a holding portion (**17**) to be sandwiched between the terminal (**13, 13**A, **14, 14**A) and the wire (**11**). The electrically conductive portion (**22**) and the holding portion (**17**) are arranged to overlap in an axial direction (X) of the wire (**11**).

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Appl. No.:	**18/008030**
Filed (or PCT Filed):	**May 21, 2021**
PCT No.:	**PCT/JP2021/019397**
PCT Pub. No.:	**WO2021/251106**
PCT Pub. Date:	December 16, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230299537 A1	Sep. 21, 2023

Foreign Application Priority Data

JP	2020-099763	Jun. 09, 2020

Publication Classification

Int. Cl.: **H01R13/52** (20060101); **H01R4/18** (20060101); **H01R13/40** (20060101)

U.S. Cl.:

CPC **H01R13/5221** (20130101); **H01R4/18** (20130101); **H01R13/40** (20130101);

Field of Classification Search

CPC: H01R (13/5221); H01R (4/18); H01R (13/40); H01R (13/5205)

USPC: 439/271

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is a national phase of PCT application No. PCT/JP2021/019397, filed on 21 May 2021, which claims priority from Japanese patent application No. 2020-099763, filed on 9 Jun. 2020, all of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION
Field of the Invention
(2) The present disclosure relates to a rubber plug unit and a connector.
Description of Related Art
(3) A connector disclosed in Patent Document 1 is provided with a wire to be held by a wire barrel portion and an insulation barrel portion of a terminal and a rubber plug (sealing body) integrally provided to the wire and the terminal to cover the insulation barrel portion.
(4) A connector disclosed in Patent Document 2 is provided with a wire to be connected to a wire connecting portion of a terminal and a rubber plug (sealing body), through which the wire is inserted.
PRIOR ART DOCUMENT
Patent Document
(5) Patent Document 1: JP 2015-090843 A Patent Document 2: JP 2017-216202 A
SUMMARY OF THE INVENTION
(6) In the case of Patent Document 1 and Patent Document 2, there is a concern that the rubber plug is shifted in an axial direction with respect to the wire. In contrast, it is also possible to adopt a configuration for restricting a position shift of the rubber plug with respect to the wire by holding the rubber plug between the insulation barrel portion of the terminal and the wire. However, if a part where the rubber plug is held by the insulation barrel portion and a part where the wire is connected by the wire barrel portion are provided side by side in the axial direction, there is a concern for the enlargement of a unit including the rubber plug and the connector.
(7) Accordingly, the present disclosure aims to provide a rubber plug unit and a connector avoiding enlargement.
(8) The present disclosure is directed to a rubber plug unit with a terminal, a wire, and a rubber plug to be held in close contact with an outer periphery of the wire, the wire including an electrically conductive portion to be connected to the terminal, the rubber plug including a holding portion to be sandwiched between the terminal and the wire, and the electrically conductive portion and the holding portion being arranged to overlap in an axial direction of the wire.
(9) According to the present disclosure, it is possible to provide a rubber plug unit and a connector avoiding enlargement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram of a connector mounted on a case in a first embodiment.
(2) FIG. **2** is an exploded side view of the rubber plug unit.
(3) FIG. **3** is a side view of the rubber plug unit.
(4) FIG. **4** is an enlarged side view of a main part of the rubber plug unit.
(5) FIG. **5** is a section along A-A of FIG. **4**.
(6) FIG. **6** is an enlarged side view of a main part of a rubber plug unit before connection of terminals in a second embodiment.
(7) FIG. **7** is an enlarged side view of the main part of the rubber plug unit.
(8) FIG. **8** is an enlarged side view of a main part of a rubber plug unit before connection of terminals in a third embodiment.
DETAILED DESCRIPTION OF THE INVENTION
(9) First, embodiments of the present disclosure are listed and described. (1) The rubber plug unit of the present disclosure is provided with a terminal, a wire, and a rubber plug to be held in close contact with an outer periphery of the wire, the wire including an electrically conductive portion to be connected to the terminal, the rubber plug including a holding portion to be sandwiched between

the terminal and the wire, and the electrically conductive portion and the holding portion being arranged to overlap in an axial direction of the wire. If the electrically conductive portion and the holding portion are arranged to overlap in the axial direction of the wire in this way, the enlargement of the rubber plug unit in the axial direction can be avoided. (2) Preferably, the wire is a single core wire. Since the single core wire is rigid to maintain a fixed shape, a state where the electrically conductive portion and the holding portion are arranged to overlap can be satisfactorily maintained. (3) The electrically conductive portion and the holding portion may be arranged side by side in a circumferential direction of the wire, and the terminal may include a collective barrel portion for collectively surrounding the electrically conductive portion and the holding portion. According to this configuration, the collective barrel portion can have both a function of being connected to the wire (function of a wire barrel portion) and a function of holding the rubber plug (function of an insulation barrel portion). As a result, an entire length of the terminal in the axial direction can be shortened as compared to a known terminal in which a wire barrel portion and an insulation barrel portion are provided side by side in the axial direction. (4) A protrusion configured to contact the electrically conductive portion may be provided on an inner periphery of the collective barrel portion. According to this configuration, the terminal can satisfactorily contact the electrically conductive portion via the protrusion. (1) A connector of the present disclosure is provided with the above rubber plug unit and a connector housing including a cavity, the wire being a single core wire, and a sealing lip to be held in close contact with an inner periphery of the cavity being provided on an outer periphery of the rubber plug. According to this configuration, liquid-tight sealing can be provided between the rubber plug and the connector housing and the inside of the wire can be maintained liquid-tight. As a result, the intrusion of oil and the like to a surface side of the connector to be connected to a mating connector can be blocked. (2) Preferably, the electrically conductive portions are arranged on both sides across the sealing lip in the axial direction of the wire, a connection terminal capable of contacting a mating terminal is connected to one electrically conductive portion, out of the electrically conductive portions arranged on the both sides, and a joint terminal coupled to a flexible coated wire is connected to the other electrically conductive portion, and the coated wire is arranged outside the connector housing. Since the flexible coated wire is arranged outside the connector housing, a degree of freedom in wire routing can be enhanced. Even if oil or the like enters the inside of the coated wire, the intrusion of the oil or the like to the connection terminal side can be blocked since liquid-tight sealing can be provided between the rubber plug and the connector housing and the inside of the wire can be maintained liquid-tight. As a result, the connection reliability of the connection terminal and the mating terminal can be ensured.

(10) Specific examples of the present disclosure are described below with reference to the drawings. Note that the present invention is not limited to these illustrations, but is represented by claims, and is intended to include all changes in the scope of claims and in the meaning and scope of equivalents.

First Embodiment

(11) A rubber plug unit **10** of a first embodiment is, for example, provided in a connector **80** mounted on a case **90** of an automatic transmission of an automotive vehicle via an O-ring **95** as shown in FIG. **1**. The connector **80** includes a connector housing **81** made of synthetic resin. The rubber plug unit **10** is provided with a wire **11**, a rubber plug **12** and terminals **13**, **14**. The terminals **13**, **14** include a connection terminal **13** and a joint terminal **14**. The rubber plug **10** is accommodated into a cavity **82** of the connector housing **81**. The connector housing **81** is connected to an unillustrated mating connector housing. The connection terminal **13** is electrically connected to an unillustrated mating terminal accommodated in the mating connector housing. Note that X, Y in each figure represent an axial direction X and a circumferential direction Y. The axial direction X corresponds to a longitudinal direction of the rubber plug unit **10** and an extending direction of the wire **11**. The circumferential direction Y is a direction about an axis and

corresponds to a direction along the outer periphery of the wire **11**. Further, in the following description, a "front side" in the axial direction X is an upper side of FIG. **1** and a left side of FIG. **2**, i.e. a side to be connected to the mating connector housing. A "rear side" in the axial direction X is opposite to the "front side" and corresponds to a lower side of FIG. **1** and a right side of FIG. **2**. In the case of the first embodiment, the connection terminal **13** is located forward of the joint terminal **14** in the axial direction X. These directions are set for the convenience of description and do not limit directions in actual use.

(12) The wire **11** is configured as a single core wire having no insulation coating. The wire **11** has a cylindrical shape and is rigid to maintain a fixed shape. In the case of the first embodiment, the wire **11** is shaped to linearly extend in the axial direction X (lateral direction of FIG. **2**).

(13) The rubber plug **12** is made of an oil-resistant rubber material. The rubber plug **12** has a hollow cylindrical shape as a whole and is arranged with an axis oriented in the axial direction X of the wire **11**. As shown in FIG. **5**, a sealing hole **15** is provided to penetrate in the axial direction X in a center of the rubber plug **12**. As shown in FIG. **2**, the rubber plug **12** includes a sealing body **16** and a pair of holding portions **17** connected to both sides in the axial direction X of the sealing body **16**.

(14) A plurality of sealing lips **18** are provided to project on the outer periphery of the sealing body **16**. The respective sealing lips **18** are provided side by side in the axial direction X on the outer periphery of the sealing body **16**. The respective sealing lips **18** are provided over the entire outer periphery of the sealing body **16**. A plurality of unillustrated inner peripheral lips are provided on the inner periphery of the sealing hole **15** of the sealing body **16**.

(15) The wire **11** is inserted into the sealing hole **15** of the rubber plug **12** in a liquid-tight manner. Each inner peripheral lip is compressed into close contact with the outer periphery of the wire **11**. In this way, liquid-tight sealing is provided between the rubber plug **12** and the wire **11**. As described above, the wire **11** is a single core wire and has no gap inside. Thus, the inside of the wire **11** is also maintained liquid-tight.

(16) As shown in FIG. **2**, the respective holding portions **17** project toward mutually opposite sides from end surfaces on both sides in the axial direction X of the sealing body **16**. The sealing hole **15** is so configured not to include inner peripheral lips on the sides of the holding portions **17** and have a hole diameter equal to or smaller than an outer diameter of the wire **11**. The holding portion **17** is formed to have a fixed thickness (thickness in a radial direction) in the axial direction X. The thickness of the holding portion **17** is smaller than a thickness of the sealing body **16** (minimum thickness in the radial direction of the sealing body **16** (e.g. a thickness of a valley part between the respective sealing lips **18** in the case of including no inner peripheral lips)).

(17) The holding portion **17** is shaped to gradually reduce a projection amount from the end surface in the axial direction X of the sealing body **16** from a lower end (one end in the radial direction) to an upper end (other end in the radial direction) of FIG. **2**. The tip of the holding portion **17** in a projecting direction is an inclined edge **19** inclined to approach the sealing body **16** from a lower end to an upper end of FIG. **2**. The holding portion **17** has a trapezoidal shape in a side view orthogonal to the axial direction X by upper and lower edges parallel to each other, a coupling edge **21** connected to the end surface in the axial direction X of the sealing body **16** and the inclined edge **19**. The respective holding portions **17** have the same shape and are shaped line-symmetrically across the sealing body **16**.

(18) As shown in FIG. **2**, in the wire **11**, electrically conductive portions **22** are exposed at positions overlapping the holding portions **17** in the axial direction X. The electrically conductive portions **22** are located above the inclined edges **19** of the holding portions **17** and, as shown in FIG. **4**, electrically connected to the terminals **13**, **14**. The electrically conductive portion **22** is a region of the outer periphery of the wire **11** and has no special shape.

(19) The holding portions **17** receive crimping forces of the terminals **13**, **14** and are held in a state sandwiched and positioned between the terminals **13**, **14** and the wire **11**.

(20) The connection terminal **13** is formed, such as by bending a metal plate material and, as shown in FIG. **2**, has an elongated shape in the axial direction X as a whole. The connection terminal **13** includes a tubular body portion **23**, a tab **24** projecting forward (leftward of FIG. **2**) in the axial direction X from the body portion **23** and a collective barrel portion **25**F connected to a rear side (right side of FIG. **2**) of the body portion **23**. The body portion **23** includes a recess-like lock receiving portion **26** to be locked by an unillustrated locking structure of the connector housing **81**.
(21) The tab **24** is connectable to the unillustrated mating terminal. The collective barrel portion **25**F includes a pair of crimping pieces **27** rising from a rear end part of a bottom wall **37** continuous from the body portion **23** while facing each other. Bead-like protrusions **28** extending in the circumferential direction Y are provided on the inner surfaces of the crimping pieces **27** as shown in FIG. **5**. As shown in FIG. **2**, recesses **29** are provided at positions corresponding to the protrusions **28** in the outer surfaces of the crimping pieces **27**. The protrusions **28** and the recesses **29** are formed by applying press-working to the outer surfaces of the crimping pieces **27**.
(22) The joint terminal **14** is formed, such as by bending a metal plate material and, as shown in FIG. **2**, has an elongated shape in the axial direction X as a whole. A length in the axial direction X of the joint terminal **14** is set shorter than that of the connection terminal **13**.
(23) The joint terminal **14** includes a collective barrel portion **25**R located on a front side in the axial direction X, a wire barrel portion **31** connected behind the collective barrel portion **25**R and an insulation barrel portion **32** located behind the wire barrel portion **31**. The collective barrel portion **25**R includes a pair of crimping pieces **27** rising from a front end part of a bottom wall **38** while facing each other. The collective barrel portion **25**R is shaped similarly to the collective barrel portion **25**F of the connection terminal **13** and includes, as shown in FIG. **4**, protrusions **28** on an inner surface and recesses **29** in an outer surface. The wire barrel portion **31** and the insulation barrel portion **32** are in the form of open barrels and, as shown in FIG. **1**, coupled to a coated wire **33**. The coated wire **33** is flexible and includes a plurality of strands and a coating surrounding the respective stands. The wire barrel portion **31** is crimped and connected to the respective strands exposed by removing the coating in an end part of the coated wire **33**. The insulation barrel portion **32** is crimped and connected to the coating in the end part of the coated wire **33**.
(24) The collective barrel portion **25**F of the connection terminal **13** collectively surrounds the holding portion **17** and the electrically conductive portions **22** on a front side in the axial direction X. The collective barrel portion **25**R of the joint terminal **14** collectively surrounds the holding portion **17** and the electrically conductive portions **22** on a rear side in the axial direction X.
(25) The collective barrel portions **25**F, **25**R are arranged in contact with or in proximity to outer peripheral parts of the end surfaces in the axial direction X of the sealing body **16**. As shown in FIG. **5**, the respective crimping pieces **27** and the bottom wall **37** are deformed into a circular shape or a flat shape close to a circular shape along the holding portion **17** and the electrically conductive portions **22** by crimping.
(26) As shown in FIG. **4**, a rear end side of the collective barrel portion **25**F and a front end side of the collective barrel portion **25**R are crimped to the holding portions **17** over an entire circumference. A central side in the axial direction X of the collective barrel portion **25**F, **25**R has a lower region **25**L to be crimped to the outer periphery of a tip side (side having the inclined edge **19**) of the corresponding holding portion **17** and an upper region **25**U to be crimped to the electrically conductive portions **22** of the wire **11** via the protrusions **28**. The protrusions **28** are in contact with the electrically conductive portions **22** of the wire **11** along the circumferential direction Y. The lower region **25**L of the collective barrel portion **25**F reduces a contact region with the holding portion **17** toward a front side. The lower region **25**L of the collective barrel portion **25**R reduces a contact region with the holding portion **17** toward a rear side.
(27) Next, the overall structures of the rubber plug unit **10** and the connector **80** of the first embodiment are described.

(28) In assembling, the wire **11** is inserted through the sealing hole **15** of the rubber plug **12**. A central side in the axial direction X of the outer periphery of the wire **11** is arranged in close contact with the inner periphery of the sealing hole **15** of the rubber plug **12**. As shown in FIG. **2**, both sides in the axial direction X of the outer periphery of the wire **11**, including the electrically conductive portions **22**, are exposed from both sides in the axial direction X of the rubber plug **12**.
(29) Subsequently, the collective barrel portions **25**F, **25**R of the terminals **13**, **14** are crimped and connected to the both sides in the axial direction X of the wire **11**. As shown in FIG. **4**, in a fixed width in the axial direction X of the collective barrel portion **25**F, **25**R, the lower region **25**L is crimped to the holding portion **17** of the rubber plug **12** and the upper region **25**U is crimped to the electrically conductive portions **22** of the wire **11** via the protrusions **28**. By crimping the lower regions **25**L of the collective barrel portions **25**F, **25**R to the holding portions **17** of the rubber plug **12**, the holding portions **17** are sandwiched and held between the collective barrel portions **25**F, **25**R and the wire **11** to restrict a position shift of the rubber plug **12** with respect to the wire **11**. By the contact of the protrusions **28** of the upper regions **25**U of the collective barrel portions **25**F, **25**R with the electrically conductive portions **22** of the wire **11**, the connection terminal **13** and the joint terminal **14** are conductively connected via the wire **11**. Further, the connection terminal **13** is conductively connected to the coated wire **33** via the joint terminal **14**.
(30) In the above way, the rubber plug unit **10** is obtained in which the wire **11**, the rubber plug **12** and the terminals **13**, **14** are integrated as shown in FIG. **3**. As shown in FIG. **1**, the rubber plug unit **10** is inserted into the cavity **82** of the connector housing **81**. Each sealing lip **18** of the rubber plug **12** is compressed into close contact with the inner periphery of the cavity **82** of the connector housing **81**. In this way, liquid-tight sealing is provided between the rubber plug **12** and the connector housing **81**. The tab **24** of the connection terminal **13** is arranged to project into a fitting space **83** (space into which the mating connector housing is fit) of the connector housing **81**. The coated wire **33** is pulled out to outside, which is a lower side of FIG. **1**, from the connector housing **81**. An end part of the coated wire **33** can be, for example, immersed in oil **70** pooled in an oil pan or the like. The oil **70** may reach the wire **11** of the rubber plug unit **10** by a capillary phenomenon along gaps between the strands of the coated wire **33**. However, in the case of the first embodiment, there is no gap inside the wire **11**, which is a single core wire, and sealing is provided between the wire **11** and the connector housing **81** by the rubber plug **12**. Thus, the oil **70** is effectively prevented from reaching the side of the connection terminal **13**. As a result, the intrusion of the oil **70** to the tab **24** of the connection terminal **13** can be avoided, and the connection reliability of the connection terminal **13** and the mating terminal can be improved.
(31) On the other hand, there is a concern that the rubber plug unit **10** becomes long in the axial direction X due to the presence of the wire **11**, which is a single core wire, and the rubber plug **12** on a central side in the axial direction X of the rubber plug unit **10**. However, since the electrically conductive portions **22** and the holding portions **17** are arranged to overlap in the axial direction X in the case of the first embodiment, a length in the axial direction X of the rubber plug unit **10** can be shortened as compared to the case where the electrically conductive portions **22** and the holing portions **17** are arranged side by side in the axial direction X. Moreover, as shown in FIG. **5**, the electrically conductive portions **22** and the holding portions **17** are arranged side by side in the circumferential direction Y of the wire **11**, and the collective barrel portions **25**F, **25**R are crimped to collectively surround the electrically conductive portions **22** and the holding portions **17**. Thus, lengths in the axial direction X of the terminals **13**, **14** can be shortened as compared to the case where a wire barrel portion and an insulation barrel portion are provided side by side in the axial direction X. As a result, the enlargement of the rubber plug unit **10** can be effectively avoided. Particularly, since the protrusions **28** to be held in contact with the electrically conductive portions **22** are provided on the inner peripheries of the crimping pieces **27** of the collective barrel portions **25**F, **25**R, a state where the collective barrel portions **25**F, **25**R are in contact with the electrically conductive portions **22** via the protrusions **28** can be satisfactorily realized.

Second Embodiment
(32) Next, a second embodiment is described using FIGS. **6** and **7**. Note that, for the sake of convenience, structures same as or corresponding to those of the first embodiment are denoted by the same reference signs in the second embodiment.
(33) In the case of the second embodiment, a rubber plug **12**A includes a sealing body **16**, a pair of holding portions **17** arranged to project toward both sides in the axial direction X of the sealing body **16** and a pair of insertion portions **34** connected to the tips of the respective holding portions **17** in projecting directions. The sealing body **16** has a hollow cylindrical shape and includes a sealing hole **15** penetrating in the axial direction X in a center. The insertion portion **34** is arranged to face at a distance from an end surface in the axial direction X of the sealing body **16**. The insertion portion **34** has an annular shape and includes an insertion hole **35** at a position coaxial with the sealing hole **15**. The holding portion **17** is provided to link an upper part of the outer periphery of the end surface in the axial direction X of the sealing body **16** and an upper part of each insertion portion **34**. The holding portion **17** has an arcuate cross-section and is formed to have a fixed thickness in the axial direction X.
(34) A connection terminal **13**A and a joint terminal **14**A include collective barrel portions **25**F, **25**R as in the first embodiment. Bead-like protrusions **28** extending in the circumferential direction Y are provided on the inner peripheries of the collective barrel portions **25**F, **25**R. Basic structures of the terminals **13**A, **14**A are similar to those of the terminals **13**, **14** of the first embodiment. However, unlike the first embodiment, the protrusions **28** and recesses **29** are formed on the sides of bottom walls **37**, **38** of the collective barrel portions **25**F, **25**R.
(35) A central side in the axial direction X of the outer periphery of a wire **11** is arranged in close contact with the inner periphery of the sealing hole **15** of the sealing body **16**. Both sides in the axial direction X of the outer periphery of the wire **11** are arranged in close contact with the inner peripheries of the insertion holes **35** of the insertion portions **34**. Electrically conductive portions **22** are provided to be exposed in parts of the outer periphery of the wire **11** between the sealing body **16** and the insertion portions **34**.
(36) As shown in FIG. **7**, the collective barrel portions **25**F, **25**R are arranged between the sealing body **16** and the insertion portions **34** in the axial direction X. Upper regions **25**U of the collective barrel portions **25**F, **25**R are arcuately deformed along the outer peripheries of the holding portions **17** by crimping and held in close contact with the outer peripheries of the holding portions **17**. The holding portions **17** are sandwiched and held between the collective barrel portions **25**F, **25**R and the wire **11**. Lower regions **25**L of the collective barrel portions **25**F, **25**R are arcuately deformed along a lower part of the outer periphery of the wire **11** by crimping, and the protrusions **28** are in close contact with the electrically conductive portions **22** of the wire **11**. The electrically conductive portions **22** are electrically connected to the terminals **13**A, **14**A via the protrusions **28**.
(37) According to the second embodiment, since the electrically conductive portions **22** and the holding portions **17** are arranged to overlap in the axial direction X and the collective barrel portions **25**F, **25**R collectively surround the electrically conductive portions **22** and the holding portions **17** as in the first embodiment according to the second embodiment, a length of a rubber plug unit **10**A in the axial direction X can be shortened.
Third Embodiment
(38) Next, a third embodiment is described using FIG. **8**. Note that, for the sake of convenience, structures same as or corresponding to those of the first embodiment are denoted by the same reference signs in the third embodiment.
(39) In the case of the third embodiment, a rubber plug **12**B includes a sealing body **16** and a pair of holding portions **17** connected to both sides in the axial direction X of the sealing body **16**. A basic structure of the rubber plug **12**B is similar to that of the rubber plug **12** of the first embodiment. The shapes of inclined edges of the respective holding portions **17** are different from those of the first embodiment. Out of the respective holding portions **17**, the holding portion **17** on

one side (left side of FIG. **8**) has an inclined edge **19** approaching the sealing body **16** from a lower end toward an upper end of FIG. **8**. The holding portion **17** on the other side (right side of FIG. **8**) has an inclined edge **19** approaching the sealing body **16** from the upper end toward the lower end of FIG. **8**. Further, a cut portion **36** along the axial direction X is formed in an upper end part (end part on a shorter side) of the holding portion **17** on the one side. A cut portion **36** along the axial direction X is also formed in a lower end part (end part on a shorter side) of the holding portion **17** on the other side. The outer periphery of a wire **11** includes electrically conductive portions **22** inside the cut portions **36**. The respective holding portions **17** are shaped point-symmetrically across the sealing body **16**.

(40) Unillustrated terminals include no parts corresponding to protrusions. The terminals are arcuately deformed along the outer peripheries of the holding portions **17** by crimping, and end parts (e.g. tip parts of crimping pieces) can enter the cut portions **36** and contact the electrically conductive portions **22** of the wire **11**. Since the electrically conductive portions **22** and the holding portions **17** are arranged at positions overlapping in the axial direction X in the third embodiment as in the first embodiment, a length of a rubber unit **10**B in the axial direction X can be shortened.

Other Embodiments

(41) The embodiments disclosed this time should be considered illustrative in all aspects, rather than restrictive.

(42) Although the wire is a single core wire having no insulation coating in the case of the first to third embodiments, the wire may be a single core wire including an insulation coating as another embodiment. If the wire is a single core wire including an insulation coating, electrically conductive portions may be provided on the outer periphery of the wire exposed by removing the insulation coating.

(43) Although the wire is a single core wire in the case of the first to third embodiments, the wire may be such a flexible coated wire that a plurality of strands are surrounded with an insulation coating as another embodiment.

(44) Although the terminals include the connection terminal and the joint terminal in the case of the first to third embodiments, only the connection terminal may be provided as another embodiment. In this case, the holding portion corresponding to the joint terminal can be omitted from the rubber plug. Further, in this case, the wire may be configured such that another connection terminal is coupled to an end part extending on a side opposite to the part to be coupled to the connection terminal.

(45) Although the terminal contacts the electrically conductive portions of the wire via the protrusions in the case of the first and second embodiments, the terminal may be configured to contact the electrically conductive portions of the wire without tip parts of the crimping pieces or the like including the protrusions as another embodiment.

LIST OF REFERENCE NUMERALS

(46) **10**, **10**A, **10**B . . . rubber plug unit **11** . . . wire **12**, **12**A, **12**B . . . rubber plug **13**, **13**A . . . connection terminal (terminal) **14**, **14**A . . . joint terminal (terminal) **15** . . . sealing hole **16** . . . sealing body **17** . . . holding portion **18** . . . sealing lip **19** . . . inclined edge **21** . . . coupling edge **22** . . . electrically conductive portion **23** . . . body portion **24** . . . tab **25**F, **25**R . . . collective barrel portion **25**L . . . lower region **25**U . . . upper region **26** . . . lock receiving portion **27** . . . crimping piece **28** . . . protrusion **29** . . . recess **31** . . . wire barrel portion **32** . . . insulation barrel portion **33** . . . coated wire **34** . . . insertion portion **35** . . . insertion hole **36** . . . cut portion **37**, **38** . . . bottom wall **70** . . . oil **80** . . . connector **81** . . . connector housing **82** . . . cavity **83** . . . fitting space **90** . . . case **95** . . . . O-ring X . . . axial direction Y . . . circumferential direction

Claims

1. A rubber plug unit, comprising: a terminal configured to be inserted in a cavity of a connector housing; a wire including an electrically conductive portion to be connected to the terminal; and a rubber plug to be held in close contact with an outer periphery of the wire, the rubber plug including a holding portion to be sandwiched between the terminal and the wire, and a sealing lip provided on an outer periphery of the rubber plug and configured to be held in close contact with an inner periphery of the cavity, the electrically conductive portion and the holding portion being arranged to overlap in an axial direction of the wire and being respectively arranged on both sides across the sealing lip in the axial direction of the wire.
2. The rubber plug unit of claim 1, wherein the wire is a single core wire.
3. The rubber plug unit of claim 1, wherein: the electrically conductive portion and the holding portion are arranged side by side in a circumferential direction of the wire, and the terminal includes a collective barrel portion for collectively surrounding the electrically conductive portion and the holding portion.
4. The rubber plug unit of claim 3, wherein each of opposite sides of the holding portion in the axial direction of the wire is formed to have an inclined edge, a protrusion is provided on an inner periphery of the collective barrel portion, and the protrusion extends in the circumferential direction of the wire and is configured to contact a portion of the electrically conductive portion extending axially from the inclined edge of the holding portion.
5. A connector, comprising: the rubber plug unit of claim 1; and the connector housing, wherein the rubber plug is inserted in the cavity of the connector housing together with the terminal.
6. The rubber plug unit of claim 2, wherein the terminal includes a connection terminal capable of contacting a mating terminal, and a joint terminal coupled to a flexible coated wire arranged outside the connector housing.
7. The rubber plug unit of claim 6, wherein the connection terminal is connected to a part of the electrically conductive portion arranged on one side of the sealing lip, and the joint terminal is connected to a part of the electrically conductive portion arranged on the other side of the sealing lip.